

Cambridge International AS Level Physics (9702)

305 Extreme-Difficulty Practice Questions

Question Paper

Pack 2

AS syllabus 2025-2027

Target series: Oct / Nov 2026

Topics 1–11

Diagrams included · 140 structured (theory) + 165 multiple-choice

Take $g = 9.81 \text{ m s}^{-2}$ and $c = 3.00 \times 10^8 \text{ m s}^{-1}$ unless stated otherwise.

Reference sheet — what you actually need (AS, topics 1–11)

Data & useful constants

Quantity	Value	Quantity	Value
acceleration of free fall, g	9.81 m s^{-2}	speed of light in vacuum, c	$3.00 \times 10^8 \text{ m s}^{-1}$
elementary charge, e	$1.60 \times 10^{-19} \text{ C}$	unified atomic mass unit, u	$1.66 \times 10^{-27} \text{ kg}$
electron mass	$9.11 \times 10^{-31} \text{ kg}$	proton mass	$1.67 \times 10^{-27} \text{ kg}$

SI base units: mass (kg), length (m), time (s), current (A), temperature (K). Every equation must be *homogeneous* — base units on both sides match. **Prefixes:** p 10^{-12} , n 10^{-9} , μ 10^{-6} , m 10^{-3} , c 10^{-2} , d 10^{-1} , k 10^3 , M 10^6 , G 10^9 , T 10^{12} . **Uncertainties:** for $y = a \pm b$ add *absolute* uncertainties; for products/quotients ($y = a^m b^n$) add *percentage* uncertainties ($\times$ the power). Precision = spread of repeats (random); accuracy = closeness to true value (systematic/zero errors).

Topic	Key relationships
Kinematics	$v = u + at$ · $s = ut + \frac{1}{2}at^2$ · $v^2 = u^2 + 2as$ · $s = \frac{1}{2}(u+v)t$. Area under v - t = displacement; gradient of v - t = acceleration; gradient of s - t = velocity. Projectiles: independent horizontal (constant v) and vertical ($a = g$) motions.
Dynamics	$F = ma$ (constant mass) = rate of change of momentum, $F = \Delta p / \Delta t$. Momentum $p = mv$. Weight $W = mg$. Conservation of momentum in all collisions; elastic $\Rightarrow$ KE conserved and relative speed of approach = relative speed of separation. Terminal velocity when drag = weight.
Forces	Moment = $F \times d$ (perp. distance). Torque of a couple = $F \times d$ (separation). Principle of moments: $\Sigma CW = \Sigma ACW$ about any point. Density $\rho = m/V$; pressure $p = F/A$; hydrostatic $\Delta p = \rho g \Delta h$; upthrust $F = \rho g V$.
Work / energy / power	$W = Fs \cos \theta$; $\Delta E_p = mg \Delta h$; $E_k = \frac{1}{2}mv^2$; efficiency = useful out / total in; $P = W/t = Fv$.
Deformation	Hooke: $F = kx$. Stress $\sigma = F/A$, strain $\epsilon = x/L$, Young modulus $E = \sigma/\epsilon$. Elastic PE = area under F - x = $\frac{1}{2}Fx = \frac{1}{2}kx^2$.
Waves	$v = f\lambda$; $T = 1/f$; intensity = power/area, $I \propto A^2$. Doppler (moving source): $f_o = f_s v / (v \pm v_s)$. Malus: $I = I_0 \cos^2 \theta$. Visible 400–700 nm; all EM waves transverse, speed c .
Superposition	Two-source: $\lambda = ax/D$. Grating: $d \sin \theta = n\lambda$. Stationary waves: nodes/antinodes, adjacent nodes $\lambda/2$ apart. Coherence = constant phase difference.
Electricity	$Q = It$; $I = Anvq$; $V = W/Q$; $R = V/I$; $P = VI = I^2 R = V^2/R$; $R = \rho L/A$.
D.C. circuits	e.m.f. $\epsilon = I(R + r)$; terminal p.d. $V = \epsilon - Ir$. Series $R = R_1 + R_2 + \dots$; parallel $1/R = 1/R_1 + 1/R_2 + \dots$. Kirchhoff I (charge), II (energy). Potential divider $V_{out} = V R_2 / (R_1 + R_2)$.
Particle physics	Nuclide ${}^A_Z X$; A & charge conserved. $\alpha = {}^4_2 \text{He}$, β^- = electron + antineutrino, β^+ = positron + neutrino, γ = photon. Quarks: u, d, s, c, t, b (u, c, t = $+\frac{2}{3}e$; d, s, b = $-\frac{1}{3}e$). p = uud, n = udd. Baryon = 3 quarks, meson = $q\bar{q}$. Leptons (e, ν) are fundamental.

β^- : $d \rightarrow u$ ($n \rightarrow p$). β^+ : $u \rightarrow d$ ($p \rightarrow n$). α -particles: discrete energies; β -particles: continuous spectrum (energy shared with the (anti)neutrino).

SECTION A — Structured (theory) questions [Q1–Q140]

1 Physical quantities and units

Q1. The drag force on a car is modelled by $F = \frac{1}{2}C_p A v^2$, where p is the air density, A the frontal area and v the speed. [7]

1. Use SI base units to show that the constant C is dimensionless.
2. A car of frontal area 2.1 m^2 travelling at 25 m s^{-1} in air of density 1.2 kg m^{-3} experiences a drag force of 560 N . Calculate C .

Q2. Estimates are an important physics skill. [6]

1. Estimate the number of times a human heart beats in a lifetime, showing your assumptions.
2. Given that a heart pumps about 70 cm^3 of blood per beat, estimate the total volume of blood pumped in a lifetime.

Q3. Express each of the following derived units in terms of SI base units only. [6]

1. the pascal (Pa)
2. the watt (W)
3. the volt (V)

Q4. A student times a simple pendulum, for which $T = 2\pi\sqrt{L/g}$. They measure the length $L = (0.994 \pm 0.002) \text{ m}$ and the time for 20 complete oscillations as $(39.8 \pm 0.4) \text{ s}$. [8]

1. Calculate a value for g .
2. Determine the percentage uncertainty in g and hence its absolute uncertainty.

Q5. In an experiment the potential difference across a resistor is $(6.0 \pm 0.1) \text{ V}$ and the current is $(0.25 \pm 0.01) \text{ A}$. [5]

1. Calculate the resistance.
2. Determine the absolute uncertainty in the resistance and state the result appropriately.

Q6. A 12 N weight hangs from a light string. A horizontal force H is applied to the weight so that the string makes an angle of 25° with the vertical and the weight is in equilibrium. [6]

1. Draw a free-body diagram and resolve forces to find the tension in the string.
2. Find the horizontal force H .

Q7. Two forces of magnitude 5.0 N and 8.0 N act at a point with an angle of 50° between them. [6]

1. Calculate the magnitude of their resultant.
 2. Find the angle between the resultant and the 8.0 N force.
-

Q8. An analogue ammeter reads 0.20 A when no current passes through it. A student then uses it to record four readings of a steady current: 2.05, 2.07, 2.06, 2.04 A. [6]

1. Name the type of error indicated by the reading of 0.20 A with no current, and state the corrected mean current.
 2. Comment on the precision of the four readings, and explain why high precision does not guarantee an accurate result here.
-

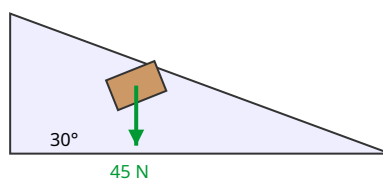
Q9. The speed v of a transverse wave on a stretched string is thought to depend on the tension T in the string and its mass per unit length μ (base units kg m^{-1}), such that $v = kT^a\mu^b$ with k a dimensionless constant. [7]

1. Use base units to determine a and b .
 2. Write down the resulting relationship.
-

Q10. Estimate the average pressure exerted on the floor by a person of mass 70 kg standing still. State clearly the estimate you make for the contact area and give your answer to an appropriate number of significant figures. [4]

Q11. A block of weight 45 N rests on a frictionless slope inclined at 30° to the horizontal. [4]

1. Resolve the weight into components parallel and perpendicular to the slope.
2. State the magnitude of the force, directed up the slope, needed to hold the block at rest.



Q12. A ball moving due east at 5.0 m s^{-1} is struck and afterwards moves due north at 5.0 m s^{-1} . [5]

1. By drawing a vector diagram, find the magnitude of the change in velocity.
 2. State the direction of this change.
-

Q13. A steel ball bearing has mass (28.2 ± 0.1) g and diameter (19.0 ± 0.1) mm.

[7]

1. Calculate the density of the steel.
 2. Determine the percentage uncertainty in the density.
-

2 Kinematics

Q14. A football is kicked from level ground with a speed of 22 m s^{-1} at 40° above the horizontal. Air resistance is negligible. [8]

1. Calculate the maximum height reached.
 2. Calculate the time of flight.
 3. Calculate the horizontal range.
-

Q15. A car starts from rest and accelerates uniformly at 2.0 m s^{-2} for 6.0 s . It then travels at constant speed for 20 s , before decelerating uniformly at 3.0 m s^{-2} to rest. [9]

1. Find the maximum speed reached.
 2. Find the total distance travelled.
 3. Find the average speed for the whole journey.
-

Q16. A ball is thrown horizontally at 15 m s^{-1} from the top of a vertical cliff of height 45 m . Air resistance is negligible. [8]

1. Calculate the time taken to reach the sea.
 2. Calculate the horizontal distance travelled.
 3. Calculate the magnitude and direction of the velocity as it hits the sea.
-

Q17. In a reaction-time test, a ruler is released from rest and a student catches it after it has fallen 0.18 m . [4]

1. Calculate the student's reaction time.
 2. State one assumption made.
-

Q18. A ball is thrown vertically upward from ground level with a speed of 25 m s^{-1} . Air resistance is negligible. [6]

1. Find the two times at which the ball is at a height of 20 m .
 2. Explain physically why there are two such times.
-

Q19. The velocity–time graph of a cyclist over 40 s consists of three straight sections: a rise from 0 to 8.0 m s^{-1} in the first 10 s, a constant 8.0 m s^{-1} for the next 20 s, and a fall from 8.0 m s^{-1} to 2.0 m s^{-1} over the final 10 s. [7]

1. Describe the motion in each of the three sections.
 2. Calculate the total distance travelled by finding the area under the graph.
-

Q20. Describe an experiment to determine the acceleration of free fall g using an electronically timed falling steel ball. Your answer should include: [6]

1. the measurements taken and the apparatus used;
 2. how a graph is used to obtain g and reduce the effect of random error.
-

Q21. A stone falls freely past two marks P and Q on a wall. It passes P and then Q, a distance 1.50 m lower, 0.100 s later. [8]

1. Find the speed of the stone at P.
 2. Find the height above P from which the stone was dropped from rest.
-

Q22. A projectile is launched from ground level at 20 m s^{-1} at 55° above the horizontal towards a vertical wall of height 4.0 m situated 6.0 m away. [7]

1. Find the time to reach the wall horizontally.
 2. Determine whether the projectile clears the top of the wall, and by how much.
-

Q23. A ball A is released from rest at the top of a tower of height 30 m. At the same instant a ball B is thrown vertically upward from the base of the tower with a speed of 15 m s^{-1} . Air resistance is negligible. [7]

1. Find the time at which the two balls are at the same height.
 2. Find that height above the ground.
-

Q24. A body starts from rest and moves with uniform acceleration 4.0 m s^{-2} in a straight line. [5]

1. Show that the distance travelled during the 5th second of its motion is 18 m.
 2. Explain why this exceeds the distance travelled in the 1st second.
-

Q25. A ball is thrown vertically upward and returns to the thrower's hand. Air resistance is negligible. [6]

1. Sketch the acceleration–time graph for the whole motion.
 2. Sketch the velocity–time graph, and mark the point corresponding to maximum height.
 3. State what feature of the velocity–time graph represents the displacement.
-

Q26. Two cars are 200 m apart on a straight road and move directly towards each other. Car X travels [5] at a constant 12 m s^{-1} ; car Y travels at a constant 8.0 m s^{-1} .

1. State the speed of X relative to Y.
 2. Find the time before they meet and the distance each has travelled.
-

3 Dynamics

Q27. A block of mass 4.0 kg is released on a slope inclined at 25° to the horizontal. A constant frictional force of 6.0 N acts on the block as it slides down. [6]

1. Calculate the resultant force on the block along the slope.
 2. Calculate its acceleration.
-

Q28. A gun fires bullets of mass 20 g horizontally at a speed of 400 m s^{-1} , at a rate of 5.0 bullets per second. [5]

1. Calculate the momentum of one bullet.
 2. Calculate the mean force needed to hold the gun steady.
-

Q29. A puck A of mass 0.50 kg moving at 4.0 m s^{-1} strikes a stationary puck B of mass 0.30 kg on a frictionless surface. After the collision A moves at 2.0 m s^{-1} at 30° to its original direction. [9]

1. Using conservation of momentum in two perpendicular directions, find the speed of B after the collision.
 2. Find the direction in which B moves.
-

Q30. Two identical trolleys each of mass m undergo a head-on elastic collision on a frictionless track. Before the collision one moves at speed u and the other is stationary. [6]

1. Using conservation of momentum and conservation of kinetic energy, show that the trolleys exchange velocities.
 2. State the relationship between relative speed of approach and relative speed of separation for any elastic collision.
-

Q31. A trolley of mass 2.0 kg moving at 3.0 m s^{-1} collides with and sticks to a stationary trolley of mass 1.0 kg. [7]

1. Calculate the common velocity after the collision.
 2. Calculate the kinetic energy lost in the collision and state what happens to this energy.
-

Q32. A resultant force acts on a stationary object of mass 5.0 kg for 4.0 s. The force–time graph is a straight line rising from 0 to 20 N over the 4.0 s. [6]

1. Explain how the impulse can be found from the graph, and calculate it.
 2. Find the final speed of the object.
-

Q33. A cannon of mass 800 kg fires a shell of mass 5.0 kg horizontally at 200 m s^{-1} .

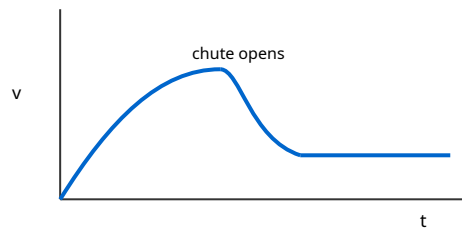
[6]

1. Calculate the recoil speed of the cannon.
 2. Compare the kinetic energy of the shell with that of the cannon, and comment.
-

Q34. A skydiver of total mass 85 kg jumps from a stationary balloon.

[7]

1. Sketch and describe the velocity–time graph from jumping until landing, referring to the forces at each stage (before and after opening the parachute).
2. State the condition for terminal velocity.



Q35. A person of mass 60 kg stands on bathroom scales inside a lift.

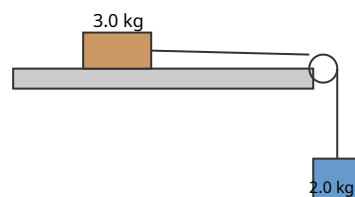
[7]

1. Find the reading (in newtons) when the lift accelerates upward at 1.5 m s^{-2} .
 2. Find the reading when the lift accelerates downward at 1.5 m s^{-2} .
 3. State the reading if the lift cable breaks and it falls freely.
-

Q36. A block of mass 3.0 kg rests on a horizontal table and is connected by a light string over a frictionless pulley at the edge to a hanging mass of 2.0 kg. A constant frictional force of 5.0 N acts on the block.

[9]

1. Calculate the acceleration of the system.
2. Calculate the tension in the string.



Q37. A ball A of mass 2.0 kg moving at 5.0 m s^{-1} makes a head-on elastic collision with a stationary ball B of mass 3.0 kg.

[8]

1. Using conservation of momentum and the elastic-collision relation (relative approach speed = relative separation speed), find the velocity of each ball after the collision.
 2. Verify that kinetic energy is conserved.
-

Q38. A ball of mass 0.15 kg travelling horizontally at 12 m s^{-1} strikes a vertical wall and rebounds horizontally at 8.0 m s^{-1} . The contact time is 0.020 s . **[7]**

1. Calculate the change in momentum of the ball.
 2. Calculate the average force exerted by the wall on the ball.
-

Q39. Two trolleys are held together with a compressed spring between them; trolley P has mass 1.5 kg and trolley Q has mass 0.50 kg . When released, Q moves off at 3.0 m s^{-1} . **[7]**

1. Find the speed of trolley P.
 2. Find the elastic potential energy that was stored in the spring.
-

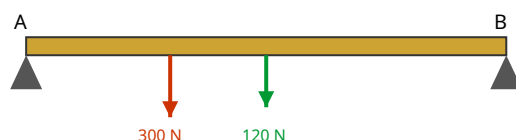
Q40. State Newton's third law of motion. **[5]**

1. A book rests on a table. Identify the force that pairs with the gravitational pull of the Earth on the book, and the force that pairs with the table's push up on the book.
 2. Explain, using Newton's laws, how the exhaust gases from a rocket propel it forward.
-

4 Forces, density and pressure

Q41. A uniform beam AB of length 5.0 m and weight 120 N rests horizontally on two supports, one at [8] each end. A load of 300 N is placed 1.5 m from end A.

1. By taking moments about A, find the force exerted by the support at B.
2. Hence find the force exerted by the support at A.



Q42. A uniform ladder of weight 200 N and length 8.0 m leans against a smooth vertical wall, making [9] an angle of 60° with the horizontal ground. The ground is rough.

1. By taking moments about the foot of the ladder, find the horizontal force exerted by the wall.
2. State the frictional force at the ground and the normal force from the ground.

Q43. Explain what is meant by a couple, and define the torque of a couple. A driver turns a steering [5] wheel of diameter 0.36 m by applying two equal, oppositely directed tangential forces of 25 N at opposite ends of a diameter.

1. Calculate the torque of the couple.
2. State why the resultant force on the wheel is zero yet it still turns.

Q44. An alloy is made by mixing 0.30 kg of metal A (density 2700 kg m^{-3}) with 0.50 kg of metal B [6] (density 8900 kg m^{-3}). Assume the volumes add.

1. Calculate the total volume of the alloy.
2. Calculate the density of the alloy.

Q45. A diver is 12 m below the surface of a lake. Atmospheric pressure is $1.01 \times 10^5 \text{ Pa}$ and the [6] density of water is 1000 kg m^{-3} .

1. Calculate the pressure due to the water alone at this depth.
2. Calculate the total pressure acting on the diver.

Q46. A U-tube contains water. Oil, which does not mix with water, is poured into the left arm. When equilibrium is reached, the oil column is 15 cm tall and its lower surface is level with the water in the right arm; the water rises 12 cm above this level on the right. [6]

1. State why the pressures at the level of the oil-water boundary are equal in the two arms.
 2. Calculate the density of the oil.
-

Q47. An object weighs 5.0 N when hung from a spring balance in air, and 3.2 N when fully immersed in water (density 1000 kg m^{-3}). [8]

1. Calculate the upthrust and hence the volume of the object.
 2. Calculate the density of the object.
-

Q48. A rectangular wooden block of base area 0.040 m^2 and height 0.20 m floats upright in water (density 1000 kg m^{-3}) with 0.15 m of its height submerged. [7]

1. Calculate the weight of the block.
 2. A load is placed on top so the block just becomes fully submerged. Calculate the weight of the load.
-

Q49. A uniform horizontal rod AB of weight 30 N and length 1.2 m is hinged at A. It is held horizontal by a light string attached at B, which makes an angle of 35° with the rod. A 50 N weight hangs from B. [8]

1. By taking moments about A, find the tension in the string.
 2. Find the vertical component of the force exerted by the hinge on the rod.
-

Q50. A meter rule is balanced horizontally on a knife edge. When a 0.50 N weight is hung at the 10 cm mark, the rule has to be moved so that it balances on the knife edge placed at the 42 cm mark. [7]

1. State where the weight of a uniform meter rule acts.
 2. By taking moments about the knife edge, find the weight of the rule.
-

Q51. A hydraulic jack has a small piston of area 2.0 cm^2 and a large piston of area 50 cm^2 . A force of 40 N is applied to the small piston. [6]

1. State the principle that makes this device work.
 2. Calculate the force produced at the large piston.
 3. Explain why this does not violate conservation of energy.
-

Q52. A load of 500 N hangs in equilibrium from a point O, held by a horizontal tie rope and a supporting cable that makes an angle of 40° with the vertical.

[7]

1. Draw a triangle of forces for point O.
 2. Calculate the tension in the cable and the tension in the horizontal tie.
-

Q53. Explain, in terms of pressure, why an object experiences an upthrust when immersed in a fluid, and state Archimedes' principle. A solid cube of side 0.10 m is fully submerged in water so that its top face is 0.20 m below the surface.

1. Calculate the difference in water pressure between the bottom and top faces.
 2. Hence calculate the upthrust, and confirm it equals $\rho g V$.
-

5 Work, energy and power

Q54. A cyclist and bicycle of total mass 90 kg free-wheel from rest down a hill, descending a vertical height of 25 m. At the bottom the speed is 18 m s^{-1} . [7]

1. Calculate the loss in gravitational potential energy.
 2. Calculate the kinetic energy at the bottom, and hence the energy dissipated by resistive forces.
-

Q55. A pump raises water from a well 8.0 m deep and delivers it at the surface through a nozzle at a speed of 6.0 m s^{-1} . Water is raised at a rate of 3.0 kg s^{-1} . [8]

1. Calculate the power needed to lift the water.
 2. Calculate the additional power needed to give the water its kinetic energy.
 3. State the minimum input power of the pump.
-

Q56. A car of mass 1200 kg accelerates from 10 m s^{-1} to 30 m s^{-1} along a level road in a distance of 250 m. A constant resistive force of 500 N acts throughout. [8]

1. Calculate the gain in kinetic energy.
 2. Calculate the work done against resistance.
 3. Hence find the average driving force provided by the engine.
-

Q57. A ski-lift carries skiers up a slope inclined at 15° to the horizontal at a steady speed of 2.5 m s^{-1} . The total mass being carried is 1800 kg and friction is negligible. [7]

1. Calculate the tension in the tow cable.
 2. Calculate the power delivered by the motor.
-

Q58. Derive the equation $P = Fv$ for the power delivered by a constant force F to an object moving at constant velocity v in the direction of the force. [6]

1. Show the derivation from the definitions of work and power.
 2. A locomotive develops a useful power of 750 kW while pulling a train at a constant 40 m s^{-1} . Find the total resistive force on the train.
-

Q59. A stone of mass 0.40 kg is projected vertically upward with an initial kinetic energy of 32 J. Air resistance does 6.0 J of work on the stone during its ascent. [6]

1. Using energy conservation, calculate the maximum height reached.
 2. State how the maximum height would compare if air resistance were negligible, without recalculating.
-

Q60. An electric motor rated at 250 W is used to raise a load of mass 30 kg vertically through 4.0 m. The operation takes 8.0 s. [6]

1. Calculate the useful output power.
 2. Calculate the efficiency of the motor.
-

Q61. A roller-coaster car of mass 500 kg passes the top of a hill at 8.0 m s^{-1} . It then descends 18 m to the lowest point of the track. A constant frictional force does $2.0 \times 10^4 \text{ J}$ of work during the descent. [6]

1. Calculate the speed of the car at the lowest point.
-

Q62. Derive the formula for kinetic energy $E_K = \frac{1}{2}mv^2$ starting from a constant resultant force F accelerating a mass m from rest to speed v . [6]

1. Give the derivation using an equation of motion and the definition of work.
 2. State the assumption about the force.
-

Q63. A wind turbine converts the kinetic energy of moving air into electrical energy. Air of density 1.2 kg m^{-3} moves at 9.0 m s^{-1} towards blades that sweep out an area of 1400 m^2 . [7]

1. Calculate the mass of air passing through the swept area each second.
 2. Calculate the maximum power available from this air.
-

Q64. A 60 kg athlete runs up a flight of stairs of total vertical height 5.0 m in 4.2 s. [5]

1. Calculate the useful power developed against gravity.
 2. Explain why the athlete's total metabolic power output is considerably greater than this value.
-

Q65. A car of mass 1000 kg has a maximum useful engine power of 42 kW. It travels up a slope inclined at 5.0° to the horizontal against a constant resistive force of 600 N. [6]

1. Calculate the maximum steady speed the car can maintain up the slope.
-

Q66. A ball of mass 0.20 kg is dropped onto a spring of force constant 800 N m^{-1} from a height of 1.2 m above the top of the spring. Assume no energy is lost. [7]

1. Calculate the speed of the ball as it first touches the spring.
 2. By equating energy, estimate the maximum compression of the spring (you may neglect the small extra fall during compression).
-

6 Deformation of solids

Q67. A steel wire of natural length 2.5 m and diameter 0.80 mm is stretched by a force of 65 N. The Young modulus of steel is 2.0×10^{11} Pa. [7]

1. Calculate the stress in the wire.
 2. Calculate the extension produced.
-

Q68. Describe an experiment to determine the Young modulus of a metal in the form of a long wire. [7]

1. State the measurements taken and the instrument used for each.
 2. Explain how a graph is used to obtain the Young modulus, and why a long thin wire is used.
-

Q69. A copper wire and a steel wire have the same length and the same diameter and are joined end to end so they hang vertically in series, supporting a load of 40 N. $E_{\text{copper}} = 1.2 \times 10^{11}$ Pa, $E_{\text{steel}} = 2.0 \times 10^{11}$ Pa. [7]

1. Explain why the two wires experience the same stress but different strains.
 2. Find the ratio of the extension of the copper wire to that of the steel wire.
-

Q70. A spring obeys Hooke's law and extends by 4.0 cm when a force of 5.0 N is applied. [7]

1. Calculate the spring constant.
 2. Calculate the elastic potential energy stored at this extension.
 3. Calculate the extra energy needed to stretch it a further 2.0 cm.
-

Q71. Two identical springs, each of spring constant 50 N m^{-1} , support a load of 12 N. [6]

1. Calculate the extension if the springs are connected in parallel (side by side).
 2. Calculate the extension if the springs are connected in series (end to end).
-

Q72. The force–extension graph for a metal wire is a straight line from the origin up to a force of 30 N at an extension of 1.5 mm (the limit of proportionality), after which it curves and the wire eventually breaks at an extension of 6.0 mm. [7]

1. Calculate the spring constant (stiffness) of the wire in its linear region.
 2. Calculate the elastic potential energy stored at the limit of proportionality.
 3. State whether the deformation at 6.0 mm is elastic or plastic, with a reason.
-

- Q73.** Define stress, strain and the Young modulus, and state the unit of each. [6]
1. Give the three definitions and units.
 2. Explain why strain has no unit.
-
- Q74.** A nylon rope of cross-sectional area $1.2 \times 10^{-4} \text{ m}^2$ and Young modulus $3.0 \times 10^9 \text{ Pa}$ hangs vertically with length 20 m. A climber of weight 700 N hangs from its lower end. [6]
1. Calculate the stress in the rope.
 2. Calculate the extension of the rope.
-
- Q75.** Distinguish between elastic and plastic deformation, and explain the meaning of the elastic limit. [6]
1. Give the distinction and define the elastic limit.
 2. State what is meant by the limit of proportionality and how it relates to Hooke's law.
-
- Q76.** A stress-strain graph is plotted for a ductile metal. It rises linearly, then reaches a peak (the ultimate tensile stress) before the metal necks and fractures. [5]
1. Explain what happens to the specimen physically after the peak of the graph.
 2. State what the area under a stress-strain graph, up to fracture, represents.
-
- Q77.** A load is applied to a wire and then removed. The loading and unloading lines on a force-extension graph do not coincide, and the unloading line returns to a non-zero extension. [5]
1. Explain what this tells you about how the wire was loaded.
 2. State what the area between the loading and unloading lines represents.
-
- Q78.** A vertical steel wire supports a mass. When the mass is 2.0 kg the extension is 0.60 mm; the wire obeys Hooke's law over this range. [6]
1. Calculate the spring constant of the wire.
 2. Predict the extension for a 3.5 kg mass, and state the assumption made.
-

7 Waves

Q79. A progressive sound wave of frequency 256 Hz travels through air at 340 m s^{-1} . [7]

1. Calculate the wavelength.
 2. Calculate the phase difference, in radians, between two points 0.44 m apart along the direction of travel.
-

Q80. A cathode-ray oscilloscope (CRO) displays the waveform of a sound. The time-base is set to 2.0 ms per division and the trace shows one complete cycle across 4.0 divisions. The y-gain is 0.50 V per division and the trace has a peak height of 3.0 divisions from the centre line. [7]

1. Calculate the frequency of the sound.
 2. State the peak voltage of the signal.
-

Q81. A point source emits sound uniformly in all directions with a power of 1.6 W. [7]

1. Calculate the intensity of the sound at a distance of 4.0 m from the source.
 2. State how the amplitude of the wave at 8.0 m compares with that at 4.0 m, justifying your answer.
-

Q82. Compare transverse and longitudinal waves. [6]

1. State the difference in terms of the direction of oscillation relative to the direction of energy transfer, giving one example of each.
 2. State which type can be polarised, and why.
-

Q83. A source of sound emits a note of frequency 480 Hz. It moves directly towards a stationary observer at 30 m s^{-1} . The speed of sound in air is 340 m s^{-1} . [7]

1. Calculate the frequency heard by the observer.
 2. State and explain qualitatively what the observer hears just after the source passes.
-

Q84. Light is plane-polarised by passing through a polarising filter. It then passes through a second filter (analyser) whose transmission axis is at 30° to that of the first. [6]

1. State Malus's law and use it to find the fraction of the intensity transmitted by the second filter.
 2. The analyser is now rotated to 90° . State the transmitted intensity and explain.
-

Q85. State the approximate range of wavelengths in free space for the principal regions of the electromagnetic spectrum. [7]

1. List, in order of increasing wavelength, the regions from γ -rays to radio waves with an approximate wavelength for each.
 2. State what all electromagnetic waves have in common in free space, and the range visible to the eye.
-

Q86. A wave travelling along a string is described by a displacement–time graph for one particle showing simple harmonic oscillation of amplitude 5.0 mm and period 0.40 s. [7]

1. State the frequency of the wave.
 2. If the wave speed is 12 m s^{-1} , calculate the wavelength.
 3. Explain the difference between the displacement–time graph for one particle and a displacement–position graph of the whole wave at one instant.
-

Q87. Unpolarised light of intensity I_0 passes through two polarising filters whose axes are parallel. [7]

1. State the intensity transmitted after the first filter (in terms of I_0).
 2. A third filter is inserted between them with its axis at 45° to both. Find the final intensity in terms of I_0 .
-

Q88. Explain what is meant by the statement that a progressive wave transfers energy. [5]

1. Describe, for a wave on the surface of water, what is transferred and what is not.
 2. State how the intensity of a progressive wave depends on its amplitude.
-

Q89. A sound source and a stationary observer are used to demonstrate the Doppler effect. The observed frequency is 1.10 times the source frequency. [6]

1. State whether the source is approaching or receding, and explain.
 2. Calculate the speed of the source (speed of sound = 340 m s^{-1}).
-

Q90. State what is meant by the polarisation of a wave, and describe how a polarising filter affects unpolarised light passing through it. [6]

1. Define polarisation.
 2. Describe an everyday application of polarising filters and the physics behind it.
-

8 Superposition

Q91. In a double-slit experiment, coherent light of wavelength 590 nm passes through two slits separated by 0.45 mm. Fringes are observed on a screen 2.2 m from the slits. [7]

1. Calculate the separation of adjacent bright fringes.
 2. State and explain what happens to the fringe separation if the slit separation is increased.
-

Q92. Explain what is meant by the terms coherence and interference, and state the conditions needed to observe a stable two-source interference pattern with light. [6]

1. Define coherence and interference.
 2. List the conditions for observable, stable fringes.
-

Q93. A diffraction grating has 500 lines per millimetre. It is illuminated normally by monochromatic light of wavelength 6.0×10^{-7} m. [9]

1. Calculate the grating spacing.
 2. Calculate the angle of diffraction of the second-order maximum.
 3. Determine the total number of bright maxima observable.
-

Q94. A stationary wave is set up on a stretched string of length 1.2 m fixed at both ends. The string vibrates in its third harmonic (three loops). [8]

1. Sketch the shape of the string, marking nodes (N) and antinodes (A).
2. Calculate the wavelength.
3. If the wave speed on the string is 240 m s^{-1} , calculate the frequency.



Q95. A tube closed at one end and open at the other has a length of 0.60 m. The speed of sound in air is 340 m s^{-1} . (Neglect end corrections.) [8]

1. Sketch the displacement pattern of the fundamental and state where the node and antinode are.
 2. Calculate the fundamental frequency.
 3. State the frequency of the next possible harmonic.
-

Q96. Microwaves from a source are directed at a metal plate, and a probe detector is moved along the line between the source and plate. The detector signal rises and falls, with minima 15 mm apart. [7]

1. Explain why a stationary wave is formed in this region.
 2. Calculate the wavelength and hence the frequency of the microwaves.
-

Q97. Explain, using the principle of superposition, how a stationary wave differs from a progressive wave. [6]

1. State the principle of superposition.
 2. Give two differences between a stationary wave and a progressive wave (energy and amplitude).
-

Q98. Light of wavelength 480 nm illuminates a double slit, giving fringes of separation 1.8 mm on a screen 1.5 m away. [7]

1. Calculate the slit separation.
 2. The blue light is replaced by red light of wavelength 660 nm. Calculate the new fringe separation.
-

Q99. Explain the meaning of diffraction, and describe an experiment using a ripple tank to show how the amount of diffraction depends on the gap width compared with the wavelength. [6]

1. Define diffraction.
 2. Describe the observations for a gap much wider than, and a gap comparable to, the wavelength.
-

Q100. Describe how a diffraction grating can be used to determine the wavelength of monochromatic light. [6]

1. State the apparatus and measurements taken.
 2. State the equation used and how the wavelength is calculated.
-

Q101. Two coherent loudspeakers face an area and produce sound of the same frequency in phase. A listener walking across the area hears alternating loud and quiet regions. [6]

1. Explain, in terms of path difference, why loud and quiet regions occur.
 2. State the path difference (in terms of wavelength λ) at a point of minimum loudness.
-

Q102. A grating spectrum is produced when white light passes through a diffraction grating. [6]

1. Explain why the central (zero-order) image is white but the higher orders are spectra.
 2. State which colour is diffracted through the largest angle in a given order, with a reason.
-

9 Electricity

Q103. A copper wire carries a current of 3.5 A. The wire has a cross-sectional area of $1.5 \times 10^{-6} \text{ m}^2$ and [7] the number density of free electrons is $8.5 \times 10^{28} \text{ m}^{-3}$.

1. Calculate the mean drift velocity of the electrons.
 2. Comment on the size of your answer and explain how a lamp can light almost instantly.
-

Q104. A charge of 90 C passes through a lamp in 1.0 minute. The lamp is connected to a 12 V supply. [7]

1. Calculate the current in the lamp.
 2. Calculate the power dissipated and the energy transferred in this time.
-

Q105. Define the resistance of a component, and state Ohm's law. [6]

1. Give both statements precisely.
 2. A component obeys $V = IR$ with $R = 15 \, \Omega$. Calculate the power dissipated when the current is 0.40 A, using two different formulae as a check.
-

Q106. A metal wire has resistance $4.0 \, \Omega$. A second wire of the same material has twice the length and [7] half the diameter of the first.

1. Explain how resistance depends on length and cross-sectional area.
 2. Calculate the resistance of the second wire.
-

Q107. Sketch and describe the I-V characteristic of a filament lamp. [6]

1. Describe the shape of the graph.
 2. Explain, in terms of the behaviour of the metal, why the resistance increases as the current increases.
-

Q108. A nichrome wire of length 1.5 m and diameter 0.30 mm has a resistivity of $1.1 \times 10^{-6} \, \Omega \text{ m}$. [7]

1. Calculate the resistance of the wire.
 2. Calculate the power dissipated when a p.d. of 6.0 V is applied across it.
-

Q109. Explain how the resistance of (a) a thermistor and (b) a light-dependent resistor (LDR) changes with external conditions, and describe one use of each. [6]

1. State the behaviour of each component.
 2. State a practical use of each.
-

Q110. An electric kettle is rated at 2.2 kW, 230 V. [8]

1. Calculate the operating current and the resistance of the element.
 2. Calculate the time taken to raise the temperature of 0.50 kg of water by 80 K, assuming all the energy heats the water. (Specific heat capacity of water = $4200 \text{ J kg}^{-1} \text{ K}^{-1}$.)
-

Q111. Two wires, X and Y, are made of the same material and have the same length. X has twice the cross-sectional area of Y. They are connected in parallel across a battery. [7]

1. Compare the resistances of X and Y.
 2. Compare the currents in, and the drift velocities of electrons in, the two wires.
-

Q112. Define the potential difference across a component in terms of energy. [5]

1. Give the definition and the defining equation.
 2. A charge of 25 mC passes through a resistor and transfers 0.30 J of energy to it. Calculate the p.d. across the resistor.
-

Q113. A student investigates how the resistance of a length of resistance wire varies and obtains a current of 0.25 A when the p.d. is 5.0 V. [6]

1. Calculate the resistance.
 2. The temperature of the wire then rises and the current falls to 0.22 A at the same p.d. Calculate the new resistance and comment on whether the wire is ohmic.
-

Q114. A semiconductor diode is connected in series with a resistor and a variable d.c. supply so that it is forward biased. [6]

1. Sketch the I-V characteristic of the diode for both forward and reverse bias.
 2. Explain the purpose of the series resistor.
-

Q115. A 12 V car battery delivers a current of 40 A to a starter motor for 3.0 s. [6]

1. Calculate the charge delivered and the energy transferred.
 2. The motor is 60% efficient. Calculate the useful mechanical energy output.
-

Q116. State what is meant by an electric current in terms of charge carriers, and explain what is meant by saying charge is quantised.

[6]

1. Define electric current and give the quantisation of charge.
 2. A current of 0.16 A flows for 5.0 s. Calculate the number of electrons that pass a point.
-

10 D.C. circuits

Q117. A battery of e.m.f. 9.0 V and internal resistance $1.5\ \Omega$ is connected to an external resistor of $6.0\ \Omega$. [7]

1. Calculate the current in the circuit.
 2. Calculate the terminal potential difference and the power dissipated inside the battery.
-

Q118. State Kirchhoff's first and second laws, and state the conservation principle underlying each. [6]

1. Give both laws and the corresponding conservation law.
 2. At a junction, currents of 2.0 A and 1.5 A flow in, and currents of 0.80 A and I flow out. Find I.
-

Q119. Three resistors of $2.0\ \Omega$, $3.0\ \Omega$ and $6.0\ \Omega$ are available. [8]

1. Calculate the combined resistance if all three are connected in series.
 2. Calculate the combined resistance if all three are connected in parallel.
 3. Calculate the combined resistance if the $3.0\ \Omega$ and $6.0\ \Omega$ are in parallel and this combination is in series with the $2.0\ \Omega$.
-

Q120. A battery of e.m.f. E and internal resistance r is connected to a variable external resistor R. [8]
When $R = 2.0\ \Omega$ the current is 2.0 A; when $R = 5.0\ \Omega$ the current is 1.0 A.

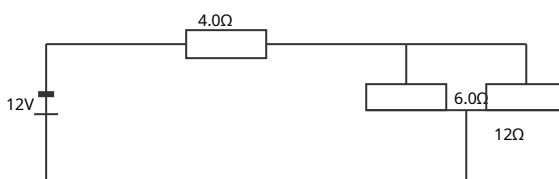
1. Write two equations and solve them for E and r.
 2. Calculate the current that would flow if the battery terminals were short-circuited.
-

Q121. Distinguish between the electromotive force (e.m.f.) of a source and the potential difference (p.d.) across its terminals. [6]

1. Define each in terms of energy per unit charge.
 2. Explain why the terminal p.d. is always less than the e.m.f. when the source supplies current, and equal to it when no current flows.
-

Q122. In the circuit shown, a 12 V battery of negligible internal resistance is connected to a $4.0\ \Omega$ resistor in series with a parallel combination of a $6.0\ \Omega$ and a $12\ \Omega$ resistor. [8]

1. Calculate the total resistance of the circuit and the current drawn from the battery.
2. Calculate the current in the $6.0\ \Omega$ resistor.



Q123. A potential divider consists of two fixed resistors, $R_1 = 3.0\ \text{k}\Omega$ and $R_2 = 9.0\ \text{k}\Omega$, connected in series across a 6.0 V supply. The output is taken across R_2 . [8]

1. Calculate the output voltage.
2. A voltmeter of resistance $18\ \text{k}\Omega$ is connected across the output. Calculate the new output voltage and comment.

Q124. Explain the principle of a potentiometer used to compare two e.m.f.s, and describe the role of the galvanometer. [7]

1. Explain how a balance point is found and what it represents.
2. A cell balances at 55.0 cm and a second cell at 43.0 cm on the same uniform wire. If the first cell has e.m.f. 1.50 V, find the e.m.f. of the second.

Q125. A light-dependent resistor (LDR) is connected in series with a $1.5\ \text{k}\Omega$ fixed resistor across a 9.0 V supply. The output is taken across the fixed resistor and fed to a circuit that switches on a lamp when the output exceeds 6.0 V. [7]

1. In darkness the LDR has a resistance of $9.0\ \text{k}\Omega$. Calculate the output voltage and state whether the lamp is on.
2. Explain what happens to the output as it becomes brighter, and hence why this arrangement switches the lamp on at night.

Q126. A network has a 6.0 V cell (negligible internal resistance) connected across two parallel branches: branch 1 is a single $10\ \Omega$ resistor; branch 2 is a $4.0\ \Omega$ and a $2.0\ \Omega$ resistor in series. [7]

1. Calculate the current in each branch.
2. Calculate the total current drawn from the cell and the total power delivered.

Q127. Two cells are connected in series with each other and with a resistor. Cell A has e.m.f. 1.5 V and [7] internal resistance $0.20\ \Omega$; cell B has e.m.f. 1.5 V and internal resistance $0.30\ \Omega$. They drive a current through an external $2.5\ \Omega$ resistor.

1. Calculate the current in the circuit.
 2. Calculate the terminal p.d. across cell B.
-

Q128. Using Kirchhoff's laws, derive the formula for the combined resistance of two resistors R_1 and [7] R_2 connected in parallel.

1. Give the derivation, stating which law is used at each step.
 2. Hence find the single resistance equivalent to $8.0\ \Omega$ and $8.0\ \Omega$ in parallel.
-

11 Particle physics

Q129. In the α -particle scattering experiment, a beam of α -particles is directed at a thin gold foil. [7]
Most pass straight through, a few are deflected through large angles, and a very small number bounce almost straight back.

1. State what each of these three observations tells us about the atom.
 2. Explain why the nucleus must be both very small and positively charged.
-

Q130. A nuclide is represented as $^{27}_{13}\text{Al}$. [6]

1. State the number of protons, neutrons and electrons in a neutral atom of this nuclide.
 2. Explain what is meant by an isotope, and give the notation for an aluminium isotope with two more neutrons.
-

Q131. Describe the composition, relative mass and relative charge of α -, β^- - and γ -radiations. [7]

1. Give the nature, mass and charge of each.
 2. State which is the most ionising and which is the most penetrating, with a brief reason.
-

Q132. A nucleus of uranium $^{238}_{92}\text{U}$ decays by emitting an α -particle. [6]

1. Write the nuclear equation, giving the nucleon and proton numbers of the daughter nucleus.
 2. Explain how nucleon number and charge are conserved in this decay.
-

Q133. During β^- decay a neutron in the nucleus changes into a proton. [6]

1. Write the equation for this change at the level of the nucleon, including the emitted particles.
 2. Explain why the emitted β -particles have a continuous range of energies.
-

Q134. State the quark composition of a proton and of a neutron. [7]

1. Give the compositions and verify the charge of each using quark charges ($u = +\frac{2}{3}e$, $d = -\frac{1}{3}e$).
 2. State the difference between a baryon and a meson.
-

Q135. During β^+ decay a proton changes into a neutron. [6]

1. Describe this change in terms of quarks.
 2. State the two particles emitted and identify them as leptons.
-

Q136. Explain what is meant by an antiparticle, and describe the positron. [5]

1. Define antiparticle and state the properties of the positron relative to the electron.
 2. State the class of fundamental particle to which the electron and neutrino belong.
-

Q137. A radium nucleus $^{226}_{88}\text{Ra}$ decays by α -emission to radon (Rn), which then decays by α -emission to polonium (Po). [6]

1. Write both decay equations, giving the nucleon and proton numbers at each stage.
-

Q138. The six flavours of quark are up, down, strange, charm, top and bottom. [6]

1. State the charge of the up, down and strange quarks, and of their antiquarks.
 2. A particle has quark composition $u\bar{u}$ (an up quark and an up antiquark). State whether it is a baryon or a meson, and calculate its charge.
-

Q139. The unified atomic mass unit (u) is used as a unit of mass in nuclear physics. [5]

1. State the approximate value of 1 u in kilograms and explain why it is a convenient unit.
 2. A carbon-12 atom has a mass of exactly 12 u. Calculate its mass in kilograms.
-

Q140. Explain why α -particles are emitted with discrete (specific) energies, whereas β -particles from a given decay have a continuous range of energies. [6]

1. Explain the α case.
 2. Explain the β case, naming the additional particle involved.
-

End of Section A — 140 structured questions.

SECTION B — Multiple choice [Q1–Q165] (choose one: A, B, C or D)

1 Physical quantities and units

1. Which of the following is a set of SI base units only?

- A kg, N, s, A
 - B m, kg, s, A
 - C m, kg, s, J
 - D m, g, s, A
-

2. The unit of the joule expressed in SI base units is:

- A kg m s^{-2}
 - B $\text{kg m}^2 \text{s}^{-2}$
 - C $\text{kg m}^2 \text{s}^{-3}$
 - D $\text{kg m}^{-1} \text{s}^{-2}$
-

3. A frequency of 2.5 GHz is equal to:

- A $2.5 \times 10^6 \text{ Hz}$
 - B $2.5 \times 10^9 \text{ Hz}$
 - C $2.5 \times 10^{12} \text{ Hz}$
 - D $2.5 \times 10^{-9} \text{ Hz}$
-

4. Which quantity is a vector?

- A kinetic energy
 - B temperature
 - C momentum
 - D electric charge
-

5. The diameter of a wire is measured as $(0.50 \pm 0.01) \text{ mm}$. The percentage uncertainty in its cross-sectional area is:

- A 2%
 - B 4%
 - C 1%
 - D 8%
-

6. Two forces of 3.0 N and 4.0 N act at right angles at a point. The magnitude of their resultant is:

- A 1.0 N
 - B 5.0 N
 - C 7.0 N
 - D 12 N
-

7. The base units $\text{kg m}^{-1} \text{s}^{-2}$ are the units of:

- A force
 - B energy
 - C pressure
 - D power
-

8. A physical quantity is estimated. Which is the best estimate of the wavelength of red light?

- A $7 \times 10^{-9} \text{ m}$
 - B $7 \times 10^{-7} \text{ m}$
 - C $7 \times 10^{-5} \text{ m}$
 - D $7 \times 10^{-3} \text{ m}$
-

9. A force of 20 N acts at 60° to the horizontal. Its horizontal component is:

- A 10 N
 - B 17 N
 - C 20 N
 - D 12 N
-

10. A micrometer has a zero error of +0.02 mm. A reading of 3.46 mm is taken. The true measurement is:

- A 3.48 mm
 - B 3.44 mm
 - C 3.46 mm
 - D 3.42 mm
-

11. Which of the following pairs contains only scalar quantities?

- A speed and energy
 - B velocity and force
 - C displacement and mass
 - D acceleration and power
-

12. The percentage uncertainties in P, Q and R are 2%, 3% and 1%. For $X = PQ/R$, the percentage uncertainty in X is:

- A 0%
 - B 2%
 - C 6%
 - D 5%
-

13. A measurement is described as accurate but not precise. This means the results are:

- A all very close together but far from the true value
 - B scattered but their average is close to the true value
 - C all identical and correct
 - D affected by a large zero error
-

14. Which equation is homogeneous with respect to units? (s = displacement, u = initial velocity, a = acceleration, t = time)

- A** $s = ut^2 + at$
 - B** $s = ut + \frac{1}{2}at^2$
 - C** $s = u + \frac{1}{2}at^2$
 - D** $s = u^2t + at$
-

15. Two forces each of magnitude 6.0 N act at a point with an angle of 120° between them. The magnitude of the resultant is:

- A** 0 N
 - B** 6.0 N
 - C** 10.4 N
 - D** 12 N
-

2 Kinematics

16. A car travels 30 m north then 40 m east. Its displacement from the start is:

- A 70 m
 - B 50 m
 - C 10 m
 - D 35 m
-

17. An object falls freely from rest. The distance it falls in the first 2.0 s is ($g = 9.8 \text{ m s}^{-2}$):

- A 9.8 m
 - B 19.6 m
 - C 39.2 m
 - D 4.9 m
-

18. The gradient of a displacement–time graph gives:

- A acceleration
 - B velocity
 - C distance
 - D force
-

19. A ball is thrown vertically up at 20 m s^{-1} . The time to reach its highest point is ($g = 10 \text{ m s}^{-2}$):

- A 1.0 s
 - B 2.0 s
 - C 4.0 s
 - D 0.5 s
-

20. The area under a velocity–time graph represents:

- A acceleration
 - B displacement
 - C speed
 - D force
-

- 21.** A projectile is launched at 30° to the horizontal at 40 m s^{-1} . Its initial vertical velocity component is:
- A** 20 m s^{-1}
 - B** 35 m s^{-1}
 - C** 40 m s^{-1}
 - D** 12 m s^{-1}
-

- 22.** A stone dropped from a bridge takes 3.0 s to reach the water. The height of the bridge is ($g = 9.8 \text{ m s}^{-2}$):
- A** 29 m
 - B** 44 m
 - C** 88 m
 - D** 15 m
-

- 23.** For a body moving with uniform acceleration, which graph is a straight horizontal line?
- A** displacement–time
 - B** velocity–time
 - C** acceleration–time
 - D** kinetic energy–time
-

- 24.** A car accelerates uniformly from 5.0 to 15 m s^{-1} in 4.0 s . Its acceleration is:
- A** 2.5 m s^{-2}
 - B** 5.0 m s^{-2}
 - C** 10 m s^{-2}
 - D** 1.25 m s^{-2}
-

- 25.** A ball is projected horizontally. Ignoring air resistance, its horizontal component of velocity during flight:
- A** increases uniformly
 - B** decreases uniformly
 - C** remains constant
 - D** first increases then decreases
-

- 26.** A car decelerates from 20 m s^{-1} to rest in 50 m . Its deceleration is:
- A** 2.0 m s^{-2}
 - B** 4.0 m s^{-2}
 - C** 8.0 m s^{-2}
 - D** 0.4 m s^{-2}
-

27. Two balls are released from the same height, one dropped and one thrown horizontally. Ignoring air resistance, they reach the ground:

- A** at the same time
 - B** the dropped one first
 - C** the thrown one first
 - D** depends on the horizontal speed
-

28. The velocity–time graph of an object is a straight line with negative gradient that crosses the time axis. At the crossing point the object:

- A** is momentarily at rest
 - B** has maximum acceleration
 - C** has maximum displacement
 - D** reverses its acceleration
-

29. A body starts from rest with uniform acceleration and travels 5.0 m in the first second. In the first two seconds it travels:

- A** 10 m
 - B** 15 m
 - C** 20 m
 - D** 25 m
-

3 Dynamics

30. A resultant force of 12 N acts on a mass of 3.0 kg. The acceleration produced is:

- A 4.0 m s^{-2}
 - B 36 m s^{-2}
 - C 0.25 m s^{-2}
 - D 9.0 m s^{-2}
-

31. The linear momentum of a 0.50 kg ball moving at 8.0 m s^{-1} is:

- A 4.0 kg m s^{-1}
 - B 16 kg m s^{-1}
 - C $0.0625 \text{ kg m s}^{-1}$
 - D 8.5 kg m s^{-1}
-

32. Force is defined as the rate of change of:

- A velocity
 - B momentum
 - C kinetic energy
 - D acceleration
-

33. A 1000 kg car experiences a resultant forward force of 2500 N. Its acceleration is:

- A 0.4 m s^{-2}
 - B 2.5 m s^{-2}
 - C 4.0 m s^{-2}
 - D 25 m s^{-2}
-

34. An object of mass 2.0 kg moving at 3.0 m s^{-1} collides with and sticks to a stationary 4.0 kg object. Their common velocity is:

- A 0.5 m s^{-1}
 - B 1.0 m s^{-1}
 - C 1.5 m s^{-1}
 - D 2.0 m s^{-1}
-

35. A ball hits a wall at 6.0 m s^{-1} and rebounds at 4.0 m s^{-1} along the same line. If its mass is 0.20 kg , the magnitude of its change in momentum is:

- A** 0.40 kg m s^{-1}
 - B** 2.0 kg m s^{-1}
 - C** 1.2 kg m s^{-1}
 - D** 0.80 kg m s^{-1}
-

36. Which is a correct statement of Newton's third law?

- A** Every action has a bigger reaction
 - B** Forces always act in pairs on the same object
 - C** If A exerts a force on B, B exerts an equal and opposite force on A
 - D** The resultant force on a body equals its mass times acceleration
-

37. A skydiver reaches terminal velocity when:

- A** the drag force is zero
 - B** the weight is zero
 - C** the drag force equals the weight
 - D** the acceleration equals g
-

38. An object explodes into two equal masses that fly apart. Compared with each other, the two pieces have:

- A** equal momenta in the same direction
 - B** equal and opposite momenta
 - C** equal kinetic energies but unequal speeds
 - D** zero total kinetic energy
-

39. A constant force of 5.0 N acts on a body for 4.0 s . The impulse delivered is:

- A** 1.25 N s
 - B** 9.0 N s
 - C** 20 N s
 - D** 0.80 N s
-

40. The weight of an object of mass 5.0 kg on a planet where $g = 3.7 \text{ m s}^{-2}$ is:

- A** 1.4 N
 - B** 8.7 N
 - C** 18.5 N
 - D** 49 N
-

41. In an elastic collision, which quantity is NOT necessarily conserved for an individual object?

- A** total momentum of the system
 - B** total kinetic energy of the system
 - C** the momentum of one object
 - D** total energy of the system
-

42. A trolley of mass m moving at speed v collides elastically head-on with an identical stationary trolley. After the collision the moving trolley:

- A** continues at speed v
 - B** stops, and the other moves off at v
 - C** rebounds at speed v
 - D** both move off at $v/2$
-

43. A jet of water of mass flow rate 2.0 kg s^{-1} hits a wall at 10 m s^{-1} and stops. The force on the wall is:

- A** 5.0 N
 - B** 20 N
 - C** 0.20 N
 - D** 100 N
-

44. The area under a force–time graph gives:

- A** work done
 - B** power
 - C** impulse
 - D** kinetic energy
-

45. A body of mass 4.0 kg on a rough surface is pulled by a 20 N force but a 8.0 N friction force opposes it. Its acceleration is:

- A** 7.0 m s^{-2}
 - B** 3.0 m s^{-2}
 - C** 5.0 m s^{-2}
 - D** 2.0 m s^{-2}
-

4 Forces, density and pressure

46. The moment of a 15 N force acting at a perpendicular distance of 0.40 m from a pivot is:

- A 6.0 N m
 - B 37.5 N m
 - C 0.027 N m
 - D 15.4 N m
-

47. A couple consists of:

- A a single force through the centre of mass
 - B two equal forces in the same direction
 - C two equal, opposite, parallel forces not in line
 - D two unequal forces at right angles
-

48. The pressure at a depth of 5.0 m in water (density 1000 kg m^{-3}) due to the water alone is ($g = 9.8 \text{ m s}^{-2}$):

- A $4.9 \times 10^3 \text{ Pa}$
 - B $4.9 \times 10^4 \text{ Pa}$
 - C $5.0 \times 10^5 \text{ Pa}$
 - D $9.8 \times 10^3 \text{ Pa}$
-

49. The density of an object of mass 240 g and volume 30 cm^3 is:

- A 8.0 g cm^{-3}
 - B 0.125 g cm^{-3}
 - C 7200 kg m^{-3}
 - D both A and its SI equivalent 8000 kg m^{-3}
-

50. A uniform beam is in equilibrium. This requires that:

- A the resultant force is zero only
 - B the resultant moment is zero only
 - C both the resultant force and the resultant moment are zero
 - D all forces act through one point
-

51. The upthrust on a body fully immersed in a fluid equals the:

- A** weight of the body
 - B** weight of fluid displaced
 - C** volume of the body
 - D** density of the fluid
-

52. A 2.0 m uniform plank of weight 60 N is pivoted at its centre. A 40 N weight hangs 0.30 m from the pivot. To balance it, a 30 N weight must hang on the other side at a distance of:

- A** 0.20 m
 - B** 0.40 m
 - C** 0.60 m
 - D** 0.90 m
-

53. A hydraulic system has pistons of area 4.0 cm^2 and 20 cm^2 . A force of 30 N on the small piston produces a force on the large piston of:

- A** 6.0 N
 - B** 30 N
 - C** 150 N
 - D** 600 N
-

54. An object floats with 80% of its volume submerged in water (1000 kg m^{-3}). Its density is:

- A** 200 kg m^{-3}
 - B** 800 kg m^{-3}
 - C** 1000 kg m^{-3}
 - D** 1250 kg m^{-3}
-

55. The centre of gravity of a body is the point where:

- A** the mass is greatest
 - B** all the weight may be taken to act
 - C** the density is uniform
 - D** the body is most stable
-

56. The torque of a couple made of two 12 N forces acting 0.25 m apart is:

- A** 1.5 N m
 - B** 3.0 N m
 - C** 6.0 N m
 - D** 48 N m
-

57. The SI base units of density are:

- A** kg m^{-3}
 - B** kg m^3
 - C** g cm^{-3}
 - D** kg m^{-2}
-

58. A manometer shows a height difference of 8.0 cm of mercury (density $1.36 \times 10^4 \text{ kg m}^{-3}$). The pressure difference is about:

- A** $1.1 \times 10^4 \text{ Pa}$
 - B** $1.1 \times 10^3 \text{ Pa}$
 - C** $1.1 \times 10^5 \text{ Pa}$
 - D** $8.0 \times 10^3 \text{ Pa}$
-

59. A ball of weight 6.0 N is held in water by a string; the upthrust is 2.5 N. The tension in the string is:

- A** 8.5 N
 - B** 3.5 N
 - C** 6.0 N
 - D** 2.5 N
-

5 Work, energy and power

60. A force of 25 N moves an object 4.0 m in the direction of the force. The work done is:

- A 6.25 J
 - B 29 J
 - C 100 J
 - D 21 J
-

61. A 2.0 kg object moves at 6.0 m s^{-1} . Its kinetic energy is:

- A 12 J
 - B 36 J
 - C 72 J
 - D 6.0 J
-

62. Power is defined as:

- A force times distance
 - B work done per unit time
 - C energy times time
 - D force per unit area
-

63. A load of mass 3.0 kg is raised 2.0 m. The gain in gravitational PE is ($g = 9.8 \text{ m s}^{-2}$):

- A 6.0 J
 - B 29 J
 - C 59 J
 - D 15 J
-

64. A machine has an efficiency of 40%. If the input energy is 500 J, the useful output is:

- A 125 J
 - B 200 J
 - C 300 J
 - D 1250 J
-

65. A car engine delivers a driving force of 800 N at a constant speed of 25 m s^{-1} . The output power is:
- A 32 W
 - B 20 kW
 - C 0.032 kW
 - D 825 W
-

66. A ball falls freely from rest. When it has fallen half of its total drop height, its speed is what fraction of its final speed?
- A $1/2$
 - B $1/\sqrt{2}$
 - C $1/4$
 - D $\sqrt{2}$
-

67. A 60 W lamp is left on for 5.0 minutes. The energy transferred is:
- A 300 J
 - B $1.8 \times 10^4 \text{ J}$
 - C 12 J
 - D $3.0 \times 10^2 \text{ J}$
-

68. A pendulum bob is released from a height of 0.20 m. Ignoring resistance, its speed at the lowest point is ($g = 9.8 \text{ m s}^{-2}$):
- A 1.4 m s^{-1}
 - B 2.0 m s^{-1}
 - C 3.9 m s^{-1}
 - D 4.0 m s^{-1}
-

69. Which statement about a body raised at constant velocity is correct?
- A Its kinetic energy increases
 - B Its potential energy increases
 - C No work is done on it
 - D Its total energy is constant
-

70. The efficiency of a system is defined as:
- A total energy input / useful output
 - B useful energy output / total energy input
 - C power output $\times$ time
 - D energy output - energy input
-

71. A crane lifts a 500 kg load at a steady 0.40 m s^{-1} . The useful power is ($g = 9.8 \text{ m s}^{-2}$):

- A** 200 W
 - B** 980 W
 - C** 1960 W
 - D** 1250 W
-

72. A 0.10 kg ball dropped from 2.0 m rebounds to 1.4 m. The energy lost in the bounce is ($g = 9.8 \text{ m s}^{-2}$):

- A** 0.59 J
 - B** 1.4 J
 - C** 2.0 J
 - D** 0.20 J
-

73. The derivation $P = Fv$ applies to a body moving:

- A** with increasing acceleration
 - B** at constant velocity with force along the motion
 - C** in a circle
 - D** against no force
-

74. A spring stores 8.0 J of elastic PE when compressed by 0.10 m. Its spring constant is:

- A** 80 N m^{-1}
 - B** 800 N m^{-1}
 - C** 1600 N m^{-1}
 - D** 160 N m^{-1}
-

6 Deformation of solids

75. Hooke's law states that, up to the limit of proportionality, the extension of a spring is:

- A inversely proportional to the load
 - B proportional to the load
 - C proportional to the square of the load
 - D independent of the load
-

76. A spring extends 5.0 cm under a load of 10 N. Its spring constant is:

- A 2.0 N m^{-1}
 - B 50 N m^{-1}
 - C 200 N m^{-1}
 - D 0.50 N m^{-1}
-

77. The Young modulus is defined as:

- A stress $\times$ strain
 - B stress / strain
 - C strain / stress
 - D force / extension
-

78. Strain has:

- A the unit pascal
 - B the unit metre
 - C no unit
 - D the unit newton
-

79. A wire of cross-sectional area $2.0 \times 10^{-7} \text{ m}^2$ carries a tension of 40 N. The stress is:

- A $2.0 \times 10^8 \text{ Pa}$
 - B $8.0 \times 10^{-6} \text{ Pa}$
 - C 20 Pa
 - D $2.0 \times 10^7 \text{ Pa}$
-

80. The elastic potential energy stored in a spring of constant k extended by x is:

- A** kx
 - B** $\frac{1}{2}kx$
 - C** $\frac{1}{2}kx^2$
 - D** kx^2
-

81. Beyond the elastic limit, a material:

- A** obeys Hooke's law
 - B** returns fully to its original length when unloaded
 - C** does not return to its original length when unloaded
 - D** has zero stress
-

82. The area under a force–extension graph represents:

- A** the spring constant
 - B** the stress
 - C** the work done in stretching
 - D** the strain
-

83. A load of 12 N stretches a wire by 0.60 mm. The energy stored (assuming Hooke's law) is:

- A** 7.2×10^{-3} J
 - B** 3.6×10^{-3} J
 - C** 7.2 J
 - D** 3.6 J
-

84. Two identical springs of constant k are joined end to end (in series). The combined spring constant is:

- A** $2k$
 - B** k
 - C** $k/2$
 - D** $4k$
-

85. The gradient of the linear part of a stress–strain graph gives the:

- A** spring constant
 - B** Young modulus
 - C** ultimate tensile stress
 - D** elastic limit
-

86. A steel wire and a copper wire have the same dimensions and carry the same load. Steel has the larger Young modulus, so the steel wire has the:

- A** larger extension
 - B** smaller extension
 - C** same extension
 - D** larger stress
-

87. A material that undergoes large plastic deformation before breaking is described as:

- A** brittle
 - B** ductile
 - C** elastic
 - D** stiff
-

7 Waves

88. A wave has frequency 200 Hz and wavelength 1.5 m. Its speed is:

- A 133 m s^{-1}
 - B 300 m s^{-1}
 - C 0.0075 m s^{-1}
 - D 201.5 m s^{-1}
-

89. The period of a wave of frequency 50 Hz is:

- A 50 s
 - B 0.02 s
 - C 0.5 s
 - D 2.0 s
-

90. A sound wave of speed 340 m s^{-1} has a frequency of 680 Hz. Its wavelength is:

- A 0.5 m
 - B 2.0 m
 - C 1.0 m
 - D 0.25 m
-

91. Which of the following is a longitudinal wave?

- A light
 - B a wave on a string
 - C sound in air
 - D a water surface wave
-

92. Which region of the electromagnetic spectrum has the longest wavelength?

- A ultraviolet
 - B X-rays
 - C radio waves
 - D infrared
-

93. Polarisation can occur only with:

- A** longitudinal waves
 - B** transverse waves
 - C** sound waves
 - D** all waves
-

94. Plane-polarised light passes through a filter whose axis is at 60° to the polarisation plane. The fraction of intensity transmitted is:

- A** 0.87
 - B** 0.75
 - C** 0.50
 - D** 0.25
-

95. If the amplitude of a progressive wave is doubled, its intensity is:

- A** halved
 - B** doubled
 - C** quadrupled
 - D** unchanged
-

96. A source of sound moves towards a stationary observer. Compared with the emitted sound, the observer hears a frequency that is:

- A** higher
 - B** lower
 - C** the same
 - D** zero
-

97. Two points on a progressive wave are separated by a quarter of a wavelength. Their phase difference is:

- A** 45°
 - B** 90°
 - C** 180°
 - D** 360°
-

98. In free space, all electromagnetic waves have the same:

- A** frequency
 - B** wavelength
 - C** speed
 - D** amplitude
-

99. On a CRO, one complete cycle of a signal spans 5.0 divisions with the time-base at 2.0 ms per division. The frequency is:

- A** 50 Hz
 - B** 100 Hz
 - C** 200 Hz
 - D** 500 Hz
-

100. The intensity of light from a point source at distance r is I . At distance $3r$ it is:

- A** $I/3$
 - B** $I/6$
 - C** $I/9$
 - D** $3I$
-

101. The approximate range of wavelengths of visible light in free space is:

- A** 40–70 nm
 - B** 400–700 nm
 - C** 4–7 μm
 - D** 4–7 mm
-

102. In a longitudinal wave, the regions where particles are furthest apart are called:

- A** compressions
 - B** rarefactions
 - C** nodes
 - D** crests
-

8 Superposition

103. In a double-slit experiment the fringe separation is given by:

- A** ax/D
 - B** $\lambda D/a$
 - C** $\lambda a/D$
 - D** aD/λ
-

104. For a diffraction grating, the maxima occur where:

- A** $a \sin\theta = n\lambda$
 - B** $d \sin\theta = n\lambda$
 - C** $d \cos\theta = n\lambda$
 - D** $d \sin\theta = \lambda/n$
-

105. Adjacent nodes on a stationary wave are separated by:

- A** λ
 - B** $\lambda/2$
 - C** $\lambda/4$
 - D** 2λ
-

106. Two sources are coherent if they have:

- A** the same amplitude only
 - B** a constant phase difference
 - C** different frequencies
 - D** the same colour only
-

107. Constructive interference occurs when the path difference is:

- A** an odd number of half-wavelengths
 - B** a whole number of wavelengths
 - C** a quarter of a wavelength
 - D** zero only
-

108. A stationary wave, compared with a progressive wave, transfers:

- A** more energy
 - B** no net energy along the wave
 - C** energy faster
 - D** energy only at nodes
-

109. A grating has 400 lines per mm. Its slit spacing is:

- A** 2.5×10^{-6} m
 - B** 4.0×10^{-4} m
 - C** 2.5×10^{-3} m
 - D** 4.0×10^5 m
-

110. Which condition is essential to observe stable two-source interference fringes?

- A** sources of different frequency
 - B** coherent sources
 - C** sources very far apart
 - D** white light only
-

111. In a double-slit experiment, moving the screen further away (increasing D) makes the fringes:

- A** closer together
 - B** further apart
 - C** unchanged
 - D** disappear
-

112. At an antinode of a stationary wave, the amplitude of oscillation is:

- A** zero
 - B** a maximum
 - C** half the maximum
 - D** equal to that at a node
-

113. Light of wavelength 5.0×10^{-7} m strikes a grating of spacing 2.0×10^{-6} m. The angle of the first-order maximum is:

- A** 7.2°
 - B** 14.5°
 - C** 30°
 - D** 45°
-

114. A string fixed at both ends vibrates in its fundamental mode. The wavelength equals:

- A** L
 - B** $2L$
 - C** $L/2$
 - D** $4L$
-

115. Light of wavelength 600 nm strikes a grating of spacing 2.0×10^{-6} m. The highest order observable is:

- A** 2
 - B** 3
 - C** 4
 - D** 5
-

116. Destructive interference between two coherent waves occurs when the path difference is:

- A** $n\lambda$
 - B** $(n + \frac{1}{2})\lambda$
 - C** $\lambda/4$
 - D** $2n\lambda$
-

9 Electricity

117. A current of 3.0 A flows for 20 s. The charge transferred is:

- A** 0.15 C
 - B** 6.7 C
 - C** 60 C
 - D** 23 C
-

118. The current in a conductor is given by $I = Anvq$. Here v represents the:

- A** speed of light
 - B** drift velocity of charge carriers
 - C** wave speed
 - D** potential difference
-

119. The potential difference across a component is defined as the:

- A** charge per unit energy
 - B** energy transferred per unit charge
 - C** power per unit current
 - D** current per unit resistance
-

120. A 12 V supply drives a current of 0.50 A through a lamp. The power dissipated is:

- A** 6.0 W
 - B** 24 W
 - C** 0.042 W
 - D** 12.5 W
-

121. The resistance of a uniform wire is given by $R = \rho L/A$. Doubling the length and doubling the area changes R by a factor of:

- A** 4
 - B** 2
 - C** 1 (unchanged)
 - D** $\frac{1}{2}$
-

122. The SI unit of resistivity is:

- A** Ω
 - B** $\Omega \text{ m}$
 - C** $\Omega \text{ m}^{-1}$
 - D** $\Omega \text{ m}^2$
-

123. As the current through a filament lamp increases, its resistance:

- A** decreases
 - B** increases
 - C** stays constant
 - D** becomes zero
-

124. The resistance of a negative-temperature-coefficient thermistor, as temperature rises:

- A** increases
 - B** decreases
 - C** stays constant
 - D** first rises then falls
-

125. The resistance of an LDR when the light intensity falling on it increases:

- A** increases
 - B** decreases
 - C** stays constant
 - D** becomes infinite
-

126. For the same current, a thinner wire (smaller area) of the same material has a drift velocity that is:

- A** smaller
 - B** larger
 - C** the same
 - D** zero
-

127. A current of 2.0 A flows through a 5.0 Ω resistor. The power dissipated is:

- A** 10 W
 - B** 20 W
 - C** 2.5 W
 - D** 40 W
-

128. A p.d. of 6.0 V is applied across a 3.0 Ω resistor. The power dissipated is:

- A** 2.0 W
 - B** 12 W
 - C** 18 W
 - D** 1.5 W
-

129. Ohm's law applies to a metallic conductor provided that:

- A** the current is very large
 - B** the temperature is constant
 - C** the p.d. is very small
 - D** it is connected in parallel
-

130. Electric current is best defined as the:

- A** number of electrons in a wire
 - B** rate of flow of charge
 - C** energy per unit charge
 - D** force on a charge
-

131. That charge is quantised means that charge:

- A** can take any value
 - B** exists only in whole multiples of e
 - C** is always negative
 - D** cannot be measured
-

132. A charge of 0.32 C passes a point. The number of electrons is ($e = 1.6 \times 10^{-19}$ C):

- A** 2.0×10^{18}
 - B** 5.0×10^{18}
 - C** 2.0×10^{19}
 - D** 5.1×10^{-20}
-

10 D.C. circuits

133. Three $6.0\ \Omega$ resistors in series have a combined resistance of:

- A $2.0\ \Omega$
 - B $6.0\ \Omega$
 - C $18\ \Omega$
 - D $0.5\ \Omega$
-

134. Two $6.0\ \Omega$ resistors in parallel have a combined resistance of:

- A $12\ \Omega$
 - B $6.0\ \Omega$
 - C $3.0\ \Omega$
 - D $1.5\ \Omega$
-

135. A cell of e.m.f. $6.0\ \text{V}$ and internal resistance $1.0\ \Omega$ is connected to a $2.0\ \Omega$ resistor. The current is:

- A $3.0\ \text{A}$
 - B $2.0\ \text{A}$
 - C $6.0\ \text{A}$
 - D $1.0\ \text{A}$
-

136. For the cell in the previous style ($E = 6.0\ \text{V}$, $r = 1.0\ \Omega$, $I = 2.0\ \text{A}$), the terminal p.d. is:

- A $6.0\ \text{V}$
 - B $4.0\ \text{V}$
 - C $2.0\ \text{V}$
 - D $8.0\ \text{V}$
-

137. Kirchhoff's first law is a consequence of the conservation of:

- A energy
 - B charge
 - C momentum
 - D mass
-

138. Kirchhoff's second law is a consequence of the conservation of:

- A** charge
 - B** energy
 - C** power
 - D** voltage
-

139. A potential divider has two equal resistors across a 10 V supply. The output across one resistor is:

- A** 10 V
 - B** 5.0 V
 - C** 2.5 V
 - D** 0 V
-

140. A potentiometer is balanced when the galvanometer reads zero. At balance, the current drawn from the cell under test is:

- A** maximum
 - B** zero
 - C** half the driver current
 - D** equal to the driver current
-

141. The 'lost volts' in a circuit with internal resistance are equal to:

- A** IR
 - B** Ir
 - C** ε
 - D** V/R
-

142. In a parallel circuit, a 6.0 V supply is connected across a $3.0\ \Omega$ resistor. The current in that resistor is:

- A** 0.5 A
 - B** 2.0 A
 - C** 18 A
 - D** 9.0 A
-

143. Two resistors, $4.0\ \Omega$ and $8.0\ \Omega$, are in series across 12 V. The p.d. across the $8.0\ \Omega$ resistor is:

- A** 4.0 V
 - B** 8.0 V
 - C** 6.0 V
 - D** 12 V
-

144. The e.m.f. of a source is the energy transferred per unit charge:

- A** to the external resistor only
 - B** in driving charge round the whole circuit
 - C** lost in the internal resistance
 - D** when no charge flows
-

145. Two identical resistors R in parallel give a combined resistance of:

- A** $2R$
 - B** R
 - C** $R/2$
 - D** $R/4$
-

146. A galvanometer is used in a null method because it:

- A** measures large currents accurately
 - B** indicates precisely when the current is zero
 - C** has a very high resistance
 - D** measures power
-

147. A cell of e.m.f. 1.5 V has internal resistance $0.30\ \Omega$. The maximum (short-circuit) current is:

- A** 5.0 A
 - B** 0.45 A
 - C** 0.20 A
 - D** 4.5 A
-

148. A $4.0\ \Omega$ resistor carries a current of 1.5 A . The power dissipated is:

- A** 6.0 W
 - B** 9.0 W
 - C** 2.7 W
 - D** 0.38 W
-

11 Particle physics

149. In the notation A_ZX , Z represents the:

- A** nucleon number
 - B** proton number
 - C** neutron number
 - D** mass in u
-

150. Isotopes of an element have the same number of:

- A** neutrons
 - B** protons
 - C** nucleons
 - D** electrons in the nucleus
-

151. An α -particle consists of:

- A** 2 protons and 2 neutrons
 - B** 1 proton and 1 neutron
 - C** 2 electrons
 - D** 2 protons only
-

152. During β^- decay, within the nucleus:

- A** a proton changes into a neutron
 - B** a neutron changes into a proton
 - C** an electron changes into a proton
 - D** a neutron is destroyed
-

153. The quark composition of a proton is:

- A** udd
 - B** uud
 - C** uds
 - D** uuu
-

154. The quark composition of a neutron is:

- A uud
 - B udd
 - C uus
 - D ddd
-

155. When $^{238}_{92}\text{U}$ emits an α -particle, the daughter nucleus is:

- A $^{234}_{90}\text{Th}$
 - B $^{238}_{90}\text{Th}$
 - C $^{234}_{92}\text{U}$
 - D $^{236}_{91}\text{Pa}$
-

156. When a nucleus ^A_ZX emits a β^- particle, the daughter nucleus is:

- A $^A_{Z+1}\text{Y}$
 - B $^A_{Z-1}\text{Y}$
 - C $^{A-1}_Z\text{Y}$
 - D $^{A-4}_{Z-2}\text{Y}$
-

157. An antiparticle has, compared with its particle, the same mass but:

- A the same charge
 - B opposite charge
 - C no mass
 - D half the charge
-

158. Which of these is a lepton?

- A proton
 - B neutron
 - C electron
 - D α -particle
-

159. A meson is composed of:

- A three quarks
 - B one quark and one antiquark
 - C two quarks
 - D three antiquarks
-

160. The charge on an up quark is:

- A** $+\frac{1}{3}e$
 - B** $+\frac{2}{3}e$
 - C** $-\frac{1}{3}e$
 - D** $-\frac{2}{3}e$
-

161. In any nuclear decay, which quantities are always conserved?

- A** mass and charge
 - B** nucleon number and charge
 - C** kinetic energy only
 - D** number of protons only
-

162. An electron antineutrino is emitted during:

- A** α -decay
 - B** γ -emission
 - C** β^- decay
 - D** β^+ decay
-

163. The α -particle scattering experiment provided evidence that the atom has a:

- A** uniform distribution of charge
 - B** small, dense, positive nucleus
 - C** negative core
 - D** solid structure
-

164. β -particles are emitted with a continuous range of energies because:

- A** they lose energy in the nucleus
 - B** a neutrino shares the energy
 - C** the nucleus recoils fully
 - D** they are not fundamental
-

165. How many flavours of quark are there?

- A** three
 - B** four
 - C** six
 - D** eight
-